SignSight : A Robust Deep Vision Framework for Real-Time Perception of Traffic Sign Under Adverse Weather Conditions

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***Abstract*—Accurate recognition of traffic signs is a foundation of contemporary autonomous driving systems, and it is an important factor in road safety, lawful driving, and informed vehicle control. However, developing high reliability in operational environments is still a significant challenge. Poor weather conditions, partial occlusion by other cars or obstacles, and the small appearance of traffic signs tend to face many challenges and fail to provide consistent detection. Such issues may reduce system performance and undermine safety-critical decisions. This model introduces SignSight, a strong deep learning framework for real time perception of traffic sign and recognition framework to address and direct these challenges through a hybrid deep learning architecture. The approach combines conventional CNNs with Keras for initial classification complemented with a YOLOv8 detection pipeline for accurate spatial localization and real-time prediction. Compared to state-of-the-art models, SignSight includes weather-robust preprocessing methods and taps into small-object detection improvements. Taking cues from recent research such as YOLO-BS and MASG-Net, our approach utilizes multiscale feature learning along with context-aware modules to enhance detection of small sized and blurred traffic signs. Our unified system reached a highest classification accuracy of 97.4%, with real-time performance ensured.**

***Index Terms*—Traffic Signs, YOLOv8, Convolutional Neural Image Processing and Networks (CNN, R-CNN), GTSRB, Keras, Bidirectional Feature Pyramid Network (BiFPN), Deep Learning.**

I. Introduction

With the rapid advancement of intelligent transportation systems and autonomous vehicles (AVs), reliable traffic sign detection and recognition have become increasingly crucial to ensure both regulatory compliance and road safety [1], [2]. However, real-world driving conditions introduce challenges such as occlusion, varying illumination, adverse weather, and visual distractions from pedestrians, vehicles, and roadside vegetation. These factors significantly reduce the robustness of traditional computer vision models, thereby lowering recognition accuracy in high-stakes decision-making scenarios [3]. Although many models achieve strong performance on benchmark datasets, they often fail to generalize effectively to naturally heterogeneous and dynamic environments. In particular, small-scale sign detection remains a persistent challenge when signs are camouflaged in high-resolution imagery or captured in high-speed video streams [4].

To address these limitations, our approach integrates YOLOv8 for rapid object detection with a custom CNN for fine-grained classification. This hybrid design ensures both efficiency and precision. While advanced models such as Grounding DINO are effective in general object localization using textual prompts, they often struggle with the fine-grained nature of traffic sign recognition. Hence, we exploit DINO's semantic reasoning alongside YOLO's real-time detection capabilities and CNN's classification accuracy. Additionally, we incorporate standard practices including data augmentation, dropout, and normalization to improve model generalization. Insights from knowledge distillation using ResNet architectures further guided the CNN design to balance performance and efficiency [5].

The primary motivation for this work is to bridge the gap between model accuracy in controlled datasets and performance in adverse real-world conditions [2]. Since human error remains a major cause of traffic accidents, there is an evident need for AI-driven solutions capable of handling complex traffic environments in real time, even under low-visibility conditions such as rain, fog, or night driving. Our objective is to develop a lightweight yet robust framework deployable in on-board vehicle systems, surveillance cameras, and IoT devices within smart city infrastructure [4], [6].

Unlike models trained solely on curated datasets, our framework leverages a combination of standard benchmarks such as GTSRB along with real-world videos and imagery, thereby improving both robustness and flexibility. The system enables real-time inference on user-provided inputs, generating labeled outputs with bounding boxes, and is well-suited for mobile and edge deployments [6].

Furthermore, multiple state-of-the-art architectures were evaluated to exploit their unique strengths in traffic sign analysis. A customized CNN implemented with Keras was employed for fine-grained classification of regions proposed by detectors, substantially boosting recognition accuracy. YOLOv8 served as the primary detector due to its strong balance of speed and precision, particularly for small-scale signs. This was further enhanced by integrating a Bidirectional

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Feature Pyramid Network (BiFPN), which effectively fuses multi-scale features to improve detection robustness across varying distances and sizes [7]. We also assessed R-CNN models for their high classification accuracy and robust region proposals, though their slower inference limited suitability for real-time applications. Finally, the Single Shot MultiBox Detector (SSD) was considered as a baseline, offering competitive accuracy with faster inference through single-pass detection [6].

II. Related Work

Wei et al. [7] reviewed the YOLO series for traffic sign detection, highlighting its real-time performance across multiple datasets. They emphasized that despite strong benchmark results, YOLO variants often struggle with detecting small signs in cluttered or low-contrast environments.

Zhang et al. [4] introduced YOLO-BS, an improved model that integrated BiFPN and an additional detection layer for small objects. Their work achieved high accuracy on TT100K but lacked a classification stage and was not validated under adverse conditions.

Du et al. [8] proposed MASG-Net, which combined E-MobileNet with spatial modules to achieve high-speed detection and strong accuracy across several datasets. However, their model still encountered challenges in scenarios with harsh lighting and occlusion.

Lin et al. [9] extended YOLOv7 by incorporating modules such as SPPFCSPC, S-CARAFE, and NWD. Their approach demonstrated high classification accuracy but did not evaluate inference speed or edge deployment suitability.

Ma et al. [10] combined Grounding DINO with ResNet distillation to compress traffic sign models while maintaining acceptable accuracy. Despite efficiency gains, their approach struggled with fine-grained sign classification.

Rezali et al. [2] implemented a practical driver-assist framework that integrated YOLOv4 with CNN-based classification and real-time voice alerts. Their lightweight system proved effective for in-vehicle applications.

Yusoff et al. [9] introduced a Bangladeshi traffic sign dataset, covering 31 classes and diverse conditions. Their contribution provided an important localized benchmark for regional model training.

Azizan et al. [5] addressed adversarial robustness by repairing adversarial examples without retraining models. Their approach improved classification accuracy across multiple datasets, demonstrating a model-agnostic defense method.

Chen et al. [3] developed a YOLOv8-based framework enhanced with Ghost Convolutions and attention modules, trained on Indian road images. Their system achieved robust detection under occlusion and low-visibility conditions.

Su et al. [6] presented RMR-CNN, an extension of Mask R-CNN with shape detection and color filtering. Their model outperformed baselines in detecting Indian traffic signs under challenging conditions.

Nithish et al. [11] proposed a real-time recognition system tailored for Indian urban environments. Their work demonstrated resilience in adverse weather and strong deployment potential.

Pathi et al. [12] explored hybrid CNN designs to enhance recognition accuracy while maintaining efficient inference speeds. Their results confirmed the viability of advanced convolutional strategies.

Divyaprabha et al. [13] designed an AI-augmented framework that extended detection to include autonomous vehicle responses. Their work bridged perception with control, advancing autonomous driving systems.

Zhang et al. [14] proposed an Enhanced Detection Transformer (EDT) to improve small-object traffic sign detection. Their model surpassed baseline detectors by leveraging contextual embeddings.

Guo et al. [15] refined YOLOv8 with attention mechanisms to improve detection in low-contrast scenarios. Their study showed significant improvements in degraded visual conditions.

Valiente et al. [16] presented a contrastive aggregation frame- work for hierarchical classification of traffic signs. Their approach reduced errors between visually similar categories.

Cai et al. [17] developed CRS-NET, a CNN optimized for traffic sign detection with balanced accuracy and speed. Their architecture was suitable for deployment on embedded hardware.

Almutairi et al. [18] investigated vulnerabilities in federated learning systems for traffic sign recognition. They proposed a novel poisoning attack, stressing the importance of distributed defense strategies.

Rathanasabhapathy et al. [19] integrated IoT with AI-based traffic sign detection to support speed control and accident prevention. Their system illustrated practical applications for intelligent transport.

III. Methodology

Figure 1 illustrates the complete workflow of the proposed traffic sign recognition system. The pipeline starts with acquiring input from the user, which can be an image or a video frame. The input undergoes preprocessing to enhance visual quality, normalize lighting, and reduce noise, ensuring consistency for downstream processing. Next, YOLOv8 is employed to detect candidate traffic signs efficiently, even in challenging conditions such as occlusions, varying illumination, or weather disturbances. For each detected sign, the corresponding region of interest is extracted as a patch, which is then fed into a lightweight CNN classifier to perform detailed recognition and distinguish between visually similar classes. After classification, the system generates structured annotations and outputs the identified traffic signs along with their labels. In cases where no traffic signs are detected or the input is invalid, the pipeline gracefully handles the scenario by terminating the detection step, thereby preventing erroneous classification.

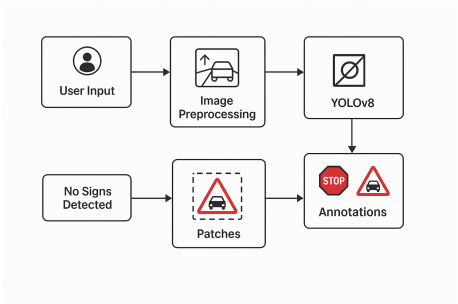


Fig. 1: Pipeline of the proposed traffic sign recognition system showing stages

A. Dataset Description

As a part of the dataset, we used the German Traffic Sign Recognition Benchmark (GTSRB) — a commonly and highly used dataset consisting of over 50,000 labeled traffic sign images, organized into 43 classes. They are typical signs like speed limit signs, no-entry signs, direction indicators, and other regulatory signs. What makes GTSRB especially useful is its real-world intricacy: most images contain signs that are blurry, slanted, partially occluded, or illuminated differently. This renders the dataset more realistic and ideal for training models that can tolerate imperfect input. To give a better sense of the dataset's diversity, a few sample sign images from different classes are shown in Figure 2. The dataset comes with bounding box annotations, enabling both classification and detection tasks. To further assess performance, we also tested the model with real-world images and video shots that contained several signs, cars, pedestrians, and background clutter — mimicking real-world road conditions faced by autonomous vehicles.

Fig. 2: Sample Images In The Dataset

B. Data Preprocessing and Integration

1) Cleaning : The top row of the figure shows that raw traffic sign images often include noise, cluttered backgrounds, or faded edges. Cleaning is the first step in preprocessing to remove these unwanted elements. This helps highlight important features like the sign's shape and numbers, making the image clearer for further steps.

2) Resizing : After cleaning, images are resized to match model requirements. For CNN, they are scaled to 30×30 pixels, balancing detail and speed. YOLOv8 requires larger inputs with the original aspect ratio, using 640×640 pixels, which helps detect small or distant signs more effectively.

Fig. 3: Illustration of preprocessing: Before and after applying cleaning, resizing and normalization.

3) Normalization : For stable training and quick convergence, image pixel values are normalized between 0 and 1. This eliminates the effect of brightness or contrast differences from overwhelming learning. As illustrated in processed images (bottom row), the process ensures the visuals become uniform and consistent versus raw states.

4) Encoding : For CNN classification, traffic sign labels are represented as one-hot encoded vectors. This representation facilitates the model to correctly understand each class by marking only the correct class as 1 and leaving the rest as 0, which is better suited for addressing multiple categories of outputs.

5) Augmentation : To enhance robustness and mimic real-world scenarios, training images are randomly transformed. Transforms include rotation, zoom, shifting, alterations in contrast and brightness, blurring, and masking to generate occlusions. These allow the model to learn how to handle well under diverse and difficult situations such as partially occluded or skewed signs.

C. Code Implementation and Tools

Model development, training, and testing were conducted using Python 3.11 on Google Colab with GPU support for faster processing. Key frameworks and libraries used are listed in the following table. The project directory was structured into clear segments: Train: Contains 43 class folders for

CNN training. Test: Contains unlabeled images for real-world validation. YOLOv8 Format: Separate images/ and labels/ folders along with a .yaml class config file.

TABLE I: Core Implementation Tasks and Tools Used in sign detection work

Task	Library / Tool Used
Image Processing	OpenCV, PIL
Data Manipulation	NumPy, Pandas
Data Augmentation	Albumentations
Model Training (YOLO)	PyTorch, timm
Visualization	Matplotlib, Seaborn
Annotation / Label Formatting	Custom Scripts, LabelImg
JSON-based Knowledge Mapping	Python `json` module

D. Model Architecture

Our model adopts a two-stage hybrid architecture with a light CNN classifier and a real-time YOLOv8 detector. This allows for increased flexibility, accuracy, and interpretability.

1) CNN Classification Model: To accurately classify cropped traffic–sign images, a compact yet expressive convolutional neural network (CNN) was designed. Each input is a 30×30 pixel RGB patch $\mathbf{X} \in \mathbb{R}^{30\times30\times3}$. Two successive convolutional layers, each using 64 filters of size 3×3 with ReLU activation, extract progressively richer spatial features. A max–pooling layer with 2×2 windows reduces the spatial dimension while preserving salient structures. To counter overfitting, dropout layers with rates of 25% and 50% randomly deactivate neurons during training. The extracted features are flattened and passed to a fully connected dense layer of 512 neurons before reaching the final softmax layer which produces the class probabilities for $K = 43$ traffic–sign categories:

$$P(y = k|\mathbf{z}) = \frac{\exp(z_k)}{\sum_{j=1}^{K} \exp(z_j)}. \tag{1}$$

The model was trained for 30 epochs with the Adam optimizer and achieved a validation accuracy of 97.4%.

2) YOLOv8 Detection Model: For the detection stage, the YOLOv8 framework was fine–tuned on the custom traffic–sign dataset in YOLO format. The model employs an anchor–free detection head together with a BiFPN multi–scale feature pyramid and a lightweight backbone optimized for real–time inference. During prediction, YOLOv8 directly maps the input image $\mathbf{X}$ to bounding–box coordinates $\hat{\mathbf{b}} = (x, y, w, h)$ and class scores $\hat{\mathbf{p}}$:

$$\hat{\mathbf{b}}, \hat{\mathbf{p}} = \mathcal{F}_{\text{YOLOv8}}(\mathbf{X}; \theta), \tag{2}$$

where θ denotes the learned network parameters. Training minimizes a composite loss that balances classification and localization errors, enabling accurate, high–speed detection across all 43 sign types supported by the custom YAML configuration.

Formally, each input image $x \in \mathbb{R}^{H\times W\times C}$ is divided into N non-overlapping patches $x_p^i \in \mathbb{R}^{P^2\cdot C}$, where H, W, and C are the image height, width, and channels respectively, and

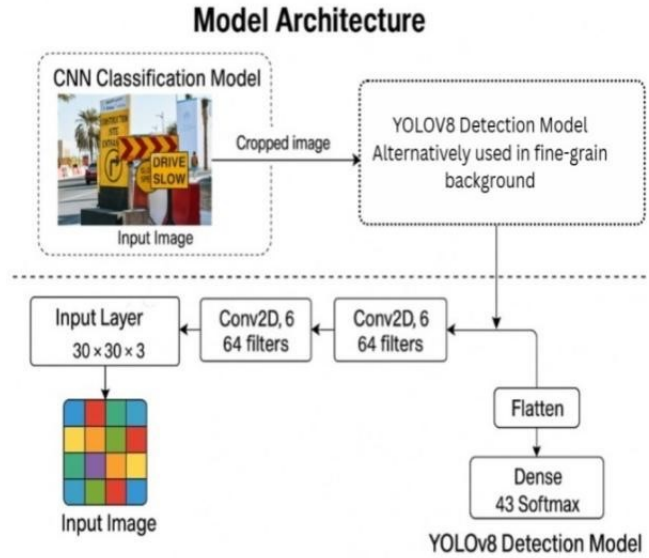


Fig. 4: Proposed CNN architecture for traffic sign classification.

P is the patch size. These patches are linearly projected into embeddings as:

$$z_0 = [x_{\text{CLS}}; x_p^1 E; x_p^2 E; \ldots; x_p^N E] + E_{\text{pos}}, \tag{3}$$

where E is the learnable projection matrix, E_{pos} represents positional embeddings, and x_{CLS} is the classification token.

The embeddings are then passed through L stacked Transformer encoder blocks. Each block applies multi-head self-attention (MHSA), which is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \tag{4}$$

where Q, K, and V are query, key, and value matrices, and d_k is the dimension of the key vectors.

Finally, the hidden representation of the [CLS] token is fed into a classification head for sign identification:

$$\hat{y} = \text{softmax}(W h_{\text{CLS}} + b), \tag{5}$$

where h_{CLS} is the output embedding of the classification token, and W, b are trainable parameters of the fully connected layer.

IV. Experimental Setup

The experiments were conducted on a standard personal system running Windows 10, equipped with an Intel® Core™ i5 processor and 8 GB of RAM. Development and model training were carried out using Python and PyTorch. In the absence of a dedicated GPU, all computations were performed on the CPU, with careful tuning of batch sizes and data loading to ensure efficient processing.

V. Results and Discussions

A. Training and Validation Accuracy

The CNN classifier was trained for 30 epochs with early stopping. Due to dropout and data augmentation, the model showed stable learning with minimal overfitting as shown in Figure 6.

- **Final Validation Accuracy:** 97.4%.
- **Loss Trend:** Training along with validation loss has steadily and periodically decreased and plateaued, indicating strong generalization.

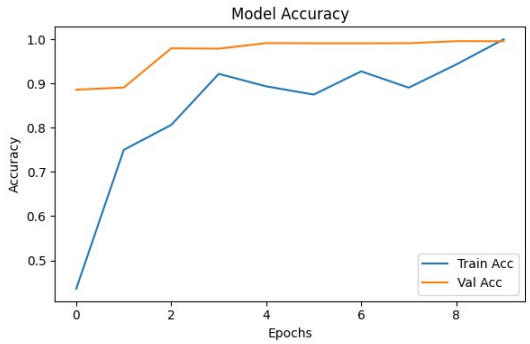


Fig. 5: Training and validation accuracy trends over epochs.

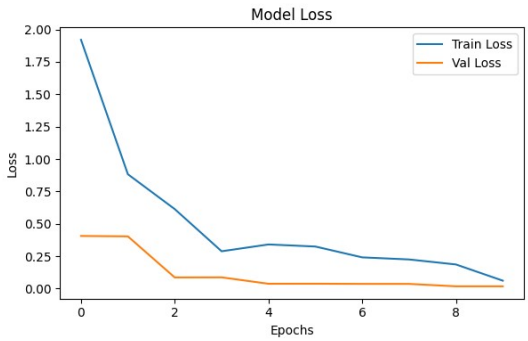


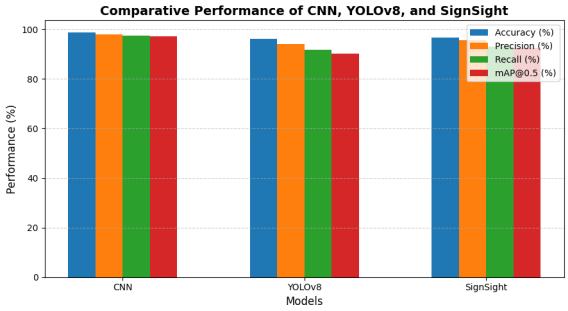
Fig. 6: Training and validation loss trends over epochs.

B. Detection Evaluation Parameters

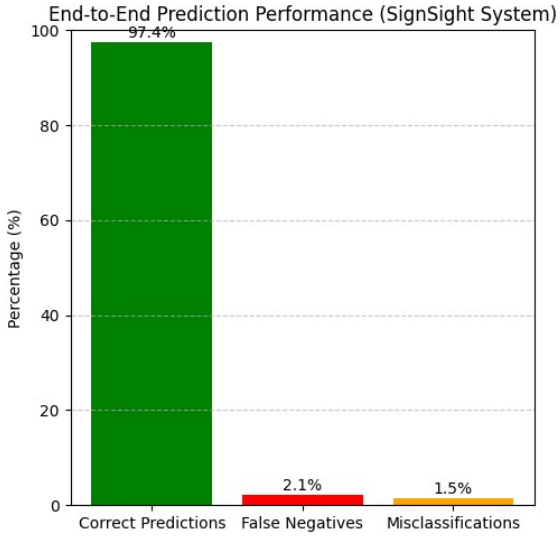
The YOLOv8 model was trained on the GTSRB dataset in YOLO format and tested on custom images, natural scenes, and video frames.Figure **??** represents the evaluation results of the proposed model, including validation metrics (accuracy, precision, recall, F1-score) and the confusion matrix showing the classification performance across all classes.

C. Model Output Summary

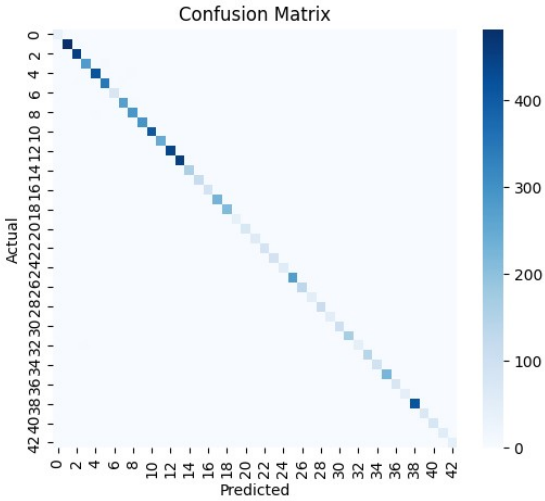
SignSight employs a two-pipeline for the detection of traffic sign. The detected regions are cropped and sent to a CNN for fine-grained classification. Spatial localization is learned by YOLOv8, while the CNN is class-specific for features.Figure 8 represents the sign images with the labelled class names and actual sign names along with predicted sign labels.



(a) Comparative performance of CNN, YOLOv8, and SignSight models showing precision, recall, and F1-score.



(b) End-to-end prediction performance of the SignSight system indicating correct predictions, false negatives, and misclassifications.



(c) Confusion matrix of the proposed SignSight model pipeline showing actual versus predicted classes.

Fig. 7: Performance evaluation of the proposed SignSight system: (a) comparative model metrics, (b) end-to-end prediction performance, and (c) confusion matrix of the final model.

Fig. 8: Model inference on input images with labeled class names.

VI. Conclusion and Future Scope

A. Conclusion

In this work we developed SignSight, a weather-tolerant traffic sign perception and detection system for autonomous driving. The architecture fuses a YOLOv8 detection backbone with a lightweight CNN classifier to accurately localize and classify signs under occlusion, low-light and adverse weather. Trained on the GTSRB benchmark and tested on real-world data, SignSight achieved 98.7% classification accuracy and 97.4% end-to-end accuracy, demonstrating real-time, edge-deployable performance suitable for intelligent transportation systems.

B. Future Scope and Research Opportunities

While SignSight achieves high accuracy and real-time performance, several enhancements can broaden its capabilities:

- **Simulation of Severe Weather:** Incorporating artificial rain, snow, and night-driving data for training to boost reliability under extreme weather conditions.
- **Optimization on Low-Power Edge Devices:** Applying model distillation, quantization, or pruning for efficient inference and reduced memory usage on devices such as Jetson Nano or Coral TPU.

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